

Preliminary

Project Name: Jukeboxd

Project members: Shresta Jha, Jack Phalen, James Shanks, Grant Moore, Shachin Chakraborty

Summary of Project:

Jukeboxd is a social music discovery platform designed to bridge the gap between "streaming" and "discussing." While modern platforms excel at music delivery, they often lack the community-driven context found in film or book communities. Jukeboxd provides a centralized hub where users can browse trending catalogs and engage with human-centric reviews.

Problem Statement and our Solution:

The Problem: Music streaming services use black-box algorithms that lack the "why" behind a recommendation. Casual listeners and enthusiasts often struggle to find authentic, peer-led reviews to guide their next listen.

The Solution: A web-based platform that combines dynamic catalog signals (Popular Albums, Singles, Artists) with a structured community review system. This allows users to move seamlessly from browsing to active participation.

Technical Architecture

To ensure scalability and ease of deployment, Jukeboxd utilizes a modern, containerized stack:

- **Backend:** Java Spring Boot (REST API)
- **Database:** MySQL (Relational data for users, music, and reviews)
- **DevOps:** Docker (Containerization for consistent environment setup)
- **Data Ingestion:** Custom scripts to populate the database via CSV datasets.

Data Source

- <https://www.kaggle.com/datasets/solomonameh/spotify-music-dataset?resource=download>

Feature Roadmap (Development Phases)

Phase 1: The Minimum Viable Product (MVP)

- **Core Logic:** Implementation of the Home Feed and Review creation.
- **Database Integration:** Establishment of MySQL via Docker and initial data ingestion.
- **Navigation:** Functional pages for Popular Albums, Singles, and Artists.

Phase 2: Persistence & Authentication

- **User Accounts:** Implementation of secure login and registration.
- **CRUD Operations:** Moving reviews from local storage to persistent database records linked to specific User and Album/Song IDs.

Phase 3: Discovery Optimization

- **Ranking Algorithms:** Developing "Popularity" metrics based on review velocity and average user ratings rather than static lists.
- **Analytics:** Backend queries to track trending media.

Phase 4: Stabilization & Deployment

- **Production Readiness:** Comprehensive README, error handling, and automated seeding scripts for one-command deployment.

The assigned Tasks/Page Ideas:

Page Ideas

- Home Page - **Shresta**
 - Title
 - List of popular albums
 - List of popular singles
 - Popular Artists
 - Recent reviews
- Profile - **James**
 - Name, General info, User Reviews, Log in, Log Out
- Listen List - **Shachin**
 - List of music you want to listen to
 - Add/remove buttons
- Review Screen - **Shresta**
 - Create a review
 - View a review
- Album View - **Grant**
 - See all songs in an album to review
 - Average album rating + number of reviews
 - Average individual song ratings
- Song View - **Grant**
 - Avg Rating
 - All reviews for song
- Top Bar - **Shachin**
 - Logo/Name
 - Genre Dropdown

- Song/Artist/Album Search bar
- User search bar

Tables - **Jack**

- Songs
 - songID (pk)
 - Title
 - ArtistID
 - AlbumID
 - Genre
- Users
 - userID (pk)
 - First name
 - Last name
 - Username
 - password
- Albums
 - albumID (pk)
 - artistId
 - Genre
 - Name
 - year
- Artists
 - artistID (pk)
 - name
- Reviews
 - userID (pk)
 - songID (pk)
 - Rating
 - comment
- ListenList
 - userID (pk)
 - songID (pk)

Songs and users -> reviews and listenList

Artist -> album and song

Album and artist -> song